

Examining the Effectiveness of Reading Questions in Introductory University Mathematics Courses

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Wabash College

- Private, independent, four year liberal arts college for men
- Grants Bachelor of Arts degrees in 21 majors
- Small (850 students)
- Student-Faculty ratio less than 10 : 1
- Selective admissions requirements

Courses

- Introductory mathematics courses
 - MAT 003 : Pre-Calculus (Axtell)
 - MAT 111 : Calculus I (Turner)
- Class meetings three times per week
- Student assessment
 - Daily homework
 - Group work / projects
 - 3 – 4 exams

Reading Questions

- Want students to read the text before class
- Many personal accounts of various techniques (Amick, 1997; Ratliff, 1997; Gold, 1998)
- Research focuses on *whether* RQs help students
 - Much research based on student evaluation responses (Boelkins & Ratliff, 2001)
 - Some research based on student exam scores (Stickles & Stickles, 2006)

Our Question

- What types of questions are most effective?

- Retention of information

- Three types of questions

Definitional: reproduce statement from text

Conceptual: more detailed; requires some thought

Computational: mimic simple example from text

- Study through reading question assignments and quizzes

Reading Question Assignments

- For every “normal” class period
- Students submit electronically evening before class
- One question of each type
- Grade 0, $\frac{1}{2}$, 1 points per problem

Example (Calculus I)

- 1 What does it mean to say a function is locally linear?
- 2 Describe how to estimate the derivative of a function by using its graph. Does this work for every function? If not, what property must the function have?
- 3 Let $f(x) = \sin(x)$. Use $x = 1.39$ and $x = 1.41$ to estimate $f'(1.4)$.

Quizzes

- 5 – 6 throughout semester
- Similar to RQs for that day — check for retention
 - Same definitional
 - Same or similar conceptual
 - Similar computational
- Additional question not in RQs — check for reading entire section

Example (Calculus I)

- 1 What does it mean to say a function is locally linear?
- 2 Describe how to estimate the derivative of a function by using its graph. Does this work for every function? If not, what property must the function have?
- 3 Let $f(x) = x^2$. Use $x = 2$ and $x = 4$ to estimate $f'(3)$.
- 4 Let $f(x) = |x|$. Does $f'(0)$ exist? Does $f'(a)$ exist if $a \neq 0$?

Axtell's Results

- MAT 003 : Pre-Calculus
- 18 students
- No-shows not counted
- Reading Question Averages — 387 student scores
 - Definitional 85.1%
 - Conceptual 66.8%
 - Computational 78.8%
- Quiz Score Averages — 6 quizzes; 98 student scores
 - Definitional 53.1%
 - Conceptual 61.7%
 - Computational 63.8%

Axtell's Score Change

- Change in score from assignment to following quiz
- Only students who did both assignment and following quiz
- 88 student scores

Definitional	- 27 pts.	-30.6%
Conceptual	+5.5 pts.	+6.25%
Computational	0 pts.	0%

Turner's Results

- MAT 111 : Calculus I
- 27 students
- No-shows not counted
- Reading Question Averages — 774 student scores
 - Definitional 90%
 - Conceptual 74%
 - Computational 60%
- Quiz Score Averages — 5 quizzes; 126 student scores
 - Definitional 72%
 - Conceptual 76%
 - Computational 54%

Turner's Score Change

- Change in score from assignment to following quiz
- Only students who did both assignment and following quiz
- 109 student scores

Definitional	– 25 pts.	–22.9%
Conceptual	– 8.5 pts.	–7.8%
Computational	– 5 pts.	–4.8%

Combined Results

- Reading Question Averages

— assumed all students answered all questions on all RQ assignments

Definitional 88.4%

Conceptual 71.6%

Computational 66.2%

- Quiz Score Averages — assumed all students took all quizzes

Definitional 63.6%

Conceptual 69.6%

Computational 58.4%

- Score Change — 197 student scores

Definitional – 52 pts. –26.4%

Conceptual – 3 pts. –1.5%

Computational – 5 pts. –2.5%

Conclusions

- Students copy definitions but do not remember them
- Students retain information conceptual and computational information, but not definitional

<http://persweb.wabash.edu/facstaff/turnerw/Presentations/SoTL-2006.pdf>

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